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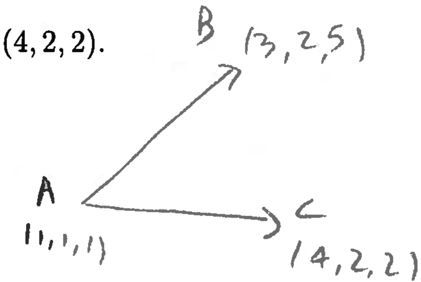
MATHEMATICS 2210

Midterm 1, Fall semester 2012

Name: KEY

15 pts **1.** Find the area of the triangle with vertices $(1, 1, 1)$, $(3, 2, 5)$ and $(4, 2, 2)$.



$\vec{AB} = (2, 1, 4)$ $\quad \vec{AC} = (3, 1, 1)$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 2 & 1 & 4 \\ 3 & 1 & 1 \end{vmatrix} = (-3, 10, -1)$$

$$\text{Area} = \frac{1}{2}\sqrt{9+100+1} = \frac{1}{2}\sqrt{110}$$

15 pts **2.** Write parametric equation for the tangent line to the curve $\vec{r}(t) = (\tan(t), 3t^2 + 1, e^{t^2})$ at $t = 0$.

$\vec{r}\,'(t) = (\sec^2(t),\ 6t,\ 2te^{t^2})$

$\vec{r}(0) = (0, 1, 1)$ $\qquad \vec{r}\,'(0) = (1, 0, 0)$

$s(t) = (0, 1, 1) + t(1, 0, 0) = (t, 1, 1)$

15 pts **3.** Find the equation of the plane containing the lines $(1,1,1) + t(1,1,2)$ and $(1,1,1) + s(2,-1,1)$.

$$\begin{vmatrix} i & j & k \\ 1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix} = (3,3,-3)$$

$$3x + 3y - 3z = D$$
$$3 + 3 - 3 = D$$
$$D = 3$$

$$\boxed{3x + 3y - 3z = 3}$$

or $x + y - z = 1$

0 pts **4.** Find κ, T, N, B, a_N, a_T for the curve $\bar{r}(t) = (\sin(3t), \cos(2t), t^2 + t)$ at $t = 0$.
(Recall that $\bar{a} = a_T T + a_N N$, $a_N = (\frac{ds}{dt})^2 \kappa$, $a_T = \frac{d^2 s}{dt^2}$, $a_T = (\bar{r}' \cdot \bar{r}'')/||\bar{r}'||$, $a_N = ||\bar{r}' \times \bar{r}''||/||\bar{r}'||$, $\kappa = ||\bar{r}' \times \bar{r}''||/||\bar{r}'||^3$, $B = T \times N$.)

$r'(t) = (3\cos 3t, -2\sin 2t, 2t+1)$ $\quad r''(t) = (-9\sin 3t, -4\cos 2t, 2)$

$r'(0) = (3, 0, 1)$ $\quad r''(0) = (0, -4, 2)$ $\quad ||r'(0)|| = \sqrt{10}$

$$\boxed{T = \frac{r'}{||r'||} = \frac{(3,0,1)}{\sqrt{10}}}$$

$$\boxed{a_T = \frac{(3,0,1)\cdot(0,-4,2)}{\sqrt{0}} = \frac{2}{\sqrt{10}}}$$

$$a_N = \frac{\left\| \begin{vmatrix} i & j & k \\ 3 & 0 & 1 \\ 0 & -4 & 2 \end{vmatrix} \right\|}{\sqrt{10}} = \frac{||(4,-6,-12)||}{\sqrt{10}} = \frac{\sqrt{196}}{\sqrt{10}} = \frac{14}{\sqrt{0}} = \frac{7\sqrt{2}}{\sqrt{5}}$$

$$\boxed{a_N = 14/\sqrt{10}}$$

$$\boxed{\kappa = \frac{14}{10\sqrt{10}}}$$

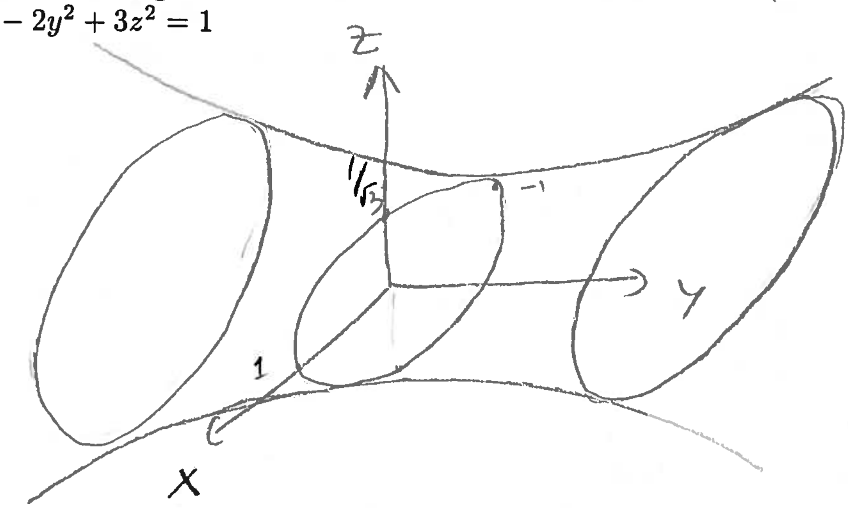
$$\boxed{N = \frac{\bar{a} - a_T T}{a_N} = \frac{(0,-4,2) - \frac{2}{\sqrt{10}}\frac{(3,0,1)}{\sqrt{10}}}{14/\sqrt{10}} = \frac{\sqrt{10}}{14}\left(-\frac{6}{10}, -4, \frac{18}{10}\right) = \frac{1}{7\sqrt{10}}(-3,-20,9)}$$

$$B = T \times N = \frac{1}{70}\begin{vmatrix} i & j & k \\ 3 & 0 & 1 \\ -3 & -20 & 9 \end{vmatrix} = \frac{1}{70}(20,-30,-60) = \left(\frac{2}{7}, \frac{-3}{7}, \frac{-6}{7}\right)$$

25 pts **5.** Graph and name the following surfaces in $\mathbb{R}^3$:

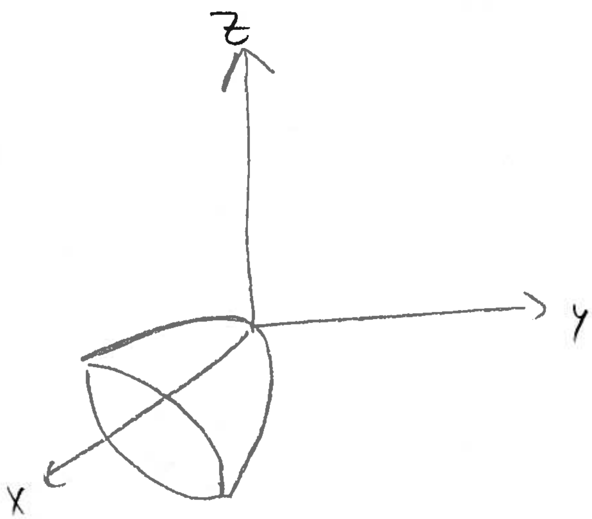
a) $x^2 - 2y^2 + 3z^2 = 1$

Hyperboloid with 1 sheet



b) $-x + y^2 + z^2 = 0$

Paraboloid



c) $x^2 - z = 0$

Cylinder